

HOFTREX 1 g with Sterilized Water For Injection 10ml

VIHOF04-var1

DESCRIPTION

Hoftrex 1g : Almost white or yellowish, crystalline, slightly hygroscopic powder.
: Pale yellow to amber solution after reconstituted with Sterile water for Injection.
Sterilized Water For Injection : Clear colorless liquid.

COMPOSITION

Each vial contains Ceftriaxone Sodium equivalent to Ceftriaxone 1g.

ACTIONS AND PHARMACOLOGY

Ceftriaxone is a broad-spectrum third generation cephalosporin antibiotic for intravenous or intramuscular administration. It binds to penicillin-binding proteins located in bacterial cytoplasmic membranes to inhibit bacterial septum and cell wall synthesis. By acylation of membrane-bound transpeptidase enzymes, ceftriaxone prevents the cross-linkage of peptidoglycan chains, which is necessary for bacterial cell wall strength and rigidity. Furthermore, cell division and growth are inhibited, as well as the elongation of susceptible bacteria, frequently causing lysis. Ceftriaxone has excellent activity against a wide spectrum of gram-negative and gram-positive bacteria. It is highly stable against hydrolysis by most beta-lactamases (penicillinases and cephalosporinases), produced by gram- negative and gram-positive bacteria.

PHARMACOKINETICS

Following the IM administration of ceftriaxone in a healthy adult, the drug appears to be completely absorbed. Peak serum concentrations are attained in between 2 to 3 hours post dosing. Ceftriaxone is reversibly bound to plasma proteins and widely distributed in body tissues and fluids. It crosses both inflamed and non-inflamed meninges, and achieves therapeutic concentrations in the cerebrospinal fluid (CSF). Ceftriaxone crosses the placenta and low concentrations have been detected in breast milk. High concentrations are found in bile. Plasma protein binding is about 85-95%. The plasma half-life of ceftriaxone is not dependent on the dose and varies between 6 and 9 hours. About 40 to 65% of a dose of ceftriaxone is excreted unchanged in the urine while the remainder is excreted in the bile and found ultimately in the faeces as an unchanged drug along with several microbiologically inactive compounds.

INDICATIONS

For treatment of the following infections caused by susceptible organisms:

- Bone and joint infections
- Uncomplicated gonorrhoea
- Intra-abdominal infections
- Lyme disease
- Meningitis
- Otitis media
- Female pelvic infections
- Respiratory tract infections, particularly bacterial pneumonia
- Bacterial septicaemia
- Skin and soft tissue infections
- Bacterial urinary tract infections

For prophylaxis of:

Post operative infections caused by susceptible organisms throughout the course of the surgical procedures.

CONTRAINDICATIONS

- This medication should not be used by patients with known allergy to penicillins, penicillin derivatives, penicillamine, or cephalosporins.
- Ceftriaxone is contraindicated in neonates ($\leq$ 28 days of age) if they require (or are expected to require) treatment with calcium- containing intravenous solutions, including calcium-containing infusions such as parenteral nutrition, because of the risk of precipitation of ceftriaxone-calcium.

PRECAUTIONS

- In order to prevent appearance of the resistant microorganisms, susceptibility should be determined and treatment should be continued only for the minimum period of time required.
- Patient history should be checked in detail and skin reaction test should be performed in order to predict adverse reaction such as shock.
- Emergency facilities should be prepared beforehand in case of the development of shock. If anaphylactic shock occurs, intravenous epinephrine has to be followed by glucocorticoid injection. After administration, patients should be kept quiet and under adequate supervision.
- It is desirable to perform laboratory test (hepatic function, renal function, blood etc.) at regular intervals during treatment.
- Shadows which have been mistaken for gallstones, but which are precipitates of calcium ceftriaxone, have been detected by sonograms. These abnormalities are commonly observed after an adult daily dose of two grams per day or more, or its equivalent in children. Rarely, these findings have been associated with symptoms. If patients develop symptoms, non-surgical management is recommended and discontinuation of ceftriaxone should be considered. The evidence suggests biliary precipitates usually disappear once ceftriaxone has been stopped.
- Pseudomembranous colitis has been reported with nearly all antibacterial agents, including ceftriaxone, and may range in severity from mild to life-threatening. Therefore, it is important to consider the diagnosis in patients who present with diarrhea subsequent to the administration of antibacterial agents.
- Prolonged use of ceftriaxone injection may result in overgrowth of non-susceptible organisms. Hence, careful observation of the patient is essential. If superinfection occurs during therapy, appropriate measures should be taken.
- Ceftriaxone injection is not reported to have adverse effect on a person's ability to drive vehicles or operate machinery.
- Use with caution if: a) Patients with history of drug allergy; b) Patients oneself of whose parents, sisters or brothers are prone to suffer from allergic symptoms such as bronchial asthma exanthema, urticaria etc. c) Patients with severe renal disorder (as the plasma concentration is maintained for a long period of time, decreased dosage or increased interval between treatments are required.) d) Patients with poor oral or parenteral nutrition patients, elderly patients, patients with poor general conditions (Cautious monitoring is required since vitamin K deficiency may occur).
- Risks and benefits of the use of ceftriaxone should be weighed in patients with colitis, gastrointestinal disease, regional enteritis or antibiotic-associated colitis as pseudomembranous colitis ranging from mild to life-threatening may occur.
- Ceftriaxone must not be mixed or administered simultaneously with calcium-containing solutions or products, even via different infusion lines.
- Calcium containing solutions or products must not be administered within 48 hours of last administration of ceftriaxone.
- Cases of fatal reactions with calcium-ceftriaxone precipitates in lung and kidneys in both term and premature neonates have been described. In some cases the infusion lines and times of administration and calcium-containing solutions differed.
- Serious and occasionally fatal hypersensitivity reactions (including anaphylactoid and severe cutaneous adverse reactions) have been reported in patients receiving therapy with beta-lactams. Before initiating therapy with Hoftrex, careful inquiry should be made concerning previous hypersensitivity reactions to penicillins, cephalosporins, carbapenems or other beta-lactam agents. If an allergic reaction occurs, Hoftrex must be discontinued immediately and appropriate alternative therapy instituted.
- In patients other than neonates, Ceftriaxone and calcium-containing solutions may be administered sequentially to one another if the infusion lines are thoroughly flushed between infusions with a compatible fluid.
- Diluents containing calcium, such as Ringer's solution or Hartmann's solution, are not to be used to reconstitute Ceftriaxone vials or to further dilute a reconstituted vial for intravenous administration because a precipitate can form. Ceftriaxone must not be administered simultaneously with calcium-containing intravenous solutions, including continuous calcium-containing infusion such as parenteral nutrition via a Y-site, because precipitation of ceftriaxone-calcium can occur.
- **Interference with laboratory test:**
 - Caution should be taken as urine-glucose test using Benedict's reagent, Fehling's reagent, Clinitest, except for Testape reaction, may give false positive results.
 - Caution should be taken as direct Coombs-test may give false positive results.
 - Ceftriaxone injection like other antibiotics, may result in false-positive tests for galactosemia.
- **Precautions on application:**
 - As vascular pain, venous thrombosis, flushing, nausea, vomiting may rarely occur by large intravenous dose, caution should be taken on preparation of injectable solution, site of injection, method of administration, etc., and injection should be given as slowly as possible (i.v. injection).
 - Ceftriaxone injection should be used immediately after reconstitution. Particularly, caution should be taken when dissolving in glutathione preparation or high concentration amino acid solution.

USE IN PREGNANCY AND LACTATION

Ceftriaxone crosses placenta. Adequate and well-controlled studies in humans have not been conducted. Ceftriaxone is excreted in breast milk, usually in low concentrations. Caution should be exercised when ceftriaxone is administered.

MAIN SIDE/ADVERSE EFFECTS

Ceftriaxone is generally well tolerated. Eosinophilia is the most frequent adverse effect. GI disturbances especially diarrhea may occur. Lesser reported GI effects include nausea, vomiting, dysgeusia and pseudomembranous colitis. Biliary sludge or pseudolithiasis due to a precipitate of calcium ceftriaxone has been seen occasionally in patients receiving ceftriaxone. Elevations of liver enzymes (SGOT and SGPT) and transient elevations of BUN have been observed. Hypersensitivity reactions (fever, swelling, itching, rash, or redness), hematological changes (leucopenia, neutropenia, thrombocytopenia, agranulocytosis, aplastic anemia or hemolytic anemia), skin reactions (erythema multiforme or Steven-Johnson syndrome) and thrombophlebitis (IV administration) are incidences that have been reported.

Nervous system disorders

Frequency 'not known': Encephalopathy*

*Reversible encephalopathy has been reported with the use of ceftriaxone, particularly when high doses are administered in patients with renal impairment and additional predisposing factors such as older age, pre-existing central nervous system disorders.

DRUG INTERACTIONS

- Ceftriaxone with Clindamycin and Verapamil: Acute verapamil toxicity, which led to complete heart block when ceftriaxone and clindamycin were administered to a patient.
- Cephalosporins with phenindione and Warfarin: Cephalosporins affected stability of anticoagulant control in a clinically significant manner.
- Probenecid has no effect on the excretion of Ceftriaxone.
- Incidence of nephrotoxicity when used concurrently with aminoglycosides has rarely been seen with other cephalosporins (except cephalothin) used at appropriate doses. The potential for increased nephrotoxicity exists when ceftriaxone is used with other nephrotoxic drug, such as loop diuretics, especially in patients with pre-existing renal function impairment.
- Concurrent use of alcohol is not recommended since it may cause disulfiram-like effects such as abdominal or stomach cramps, nausea, vomiting, headache, hypotension, palpitations, shortness of breath, tachycardia, sweating.
- The admixture of beta-lactam antibacterials (penicillins and cephalosporins) and aminoglycosides may result in substantial mutual inactivation. If they are administered concurrently, they should be administered in separate sites. Do not mix them in the same intravenous bag or bottle.

OVERDOSE

There is no known specific antidote to ceftriaxone. Drug concentration would not be reduced by hemodialysis or peritoneal dialysis. Treatment of overdosage is generally symptomatic and supportive.

DOSAGE AND ADMINISTRATION

Hovid Hoftrex may be administered by deep intramuscular injection, slow intravenous injection, or as a slow intravenous infusion, after reconstitution of the solution according to the directions given below. Dosage and mode of administration should be determined by the severity of the infection, susceptibility of the causative organism and the patient's condition. Under most circumstances a once- daily dose - or, in the specified indications, a single dose - will give satisfactory therapeutic results.

Adults and children 12 years and over:

Standard therapeutic dosage: 1 g once daily.

Severe infections: 2 - 4 g daily, normally as a single dose every 24 hours. Dose is given as single intramuscular dose.

The duration of therapy varies according to the course of the disease. As with antibiotic therapy in general, administration of hovid Hoftrex should be continued for a minimum of 48 to 72 hours after the patient has become a febrile or evidence of bacterial eradication has been obtained.

Acute, uncomplicated gonorrhea: A single dose of 250 mg intramuscularly should be administered. Simultaneous administration of probenecid is not indicated.

Meningitis: Initial therapeutic dose be 100 mg/kg (not to exceed 4 grams). Thereafter, a total daily dose of 100 mg/kg/day (not to exceed 4 grams daily) is recommended. Dose is given as single intramuscular dose.

The daily dose may be administered once a day (or unequally divided doses every 12 hours). The usual duration of therapy is 7 to 14 days.

Peri-operative prophylaxis: Usually 1g as a single intramuscular or slow intravenous dose, 30-90 minutes prior to surgery to prevent postoperative infection surgery

Elderly:

These dosages do not require modification in elderly patients provided that renal and hepatic function is satisfactory.

Neonates, infants and children up to 12 years:

The following dosage schedules are recommended for once daily administration:

Neonates

A daily dose of 20 – 50 mg/kg body weight, not to exceed 50 mg/kg. In the neonate, the intravenous dose should be given over 60 minutes to reduce the displacement of bilirubin from albumin, thereby reducing the potential risk of bilirubin encephalopathy.

Infants and children of up to 12 years

Standard therapeutic dosage: 20 – 50 mg/kg body weight once daily.

In severe infections up to 80mg/kg body weight daily may be given. For children with body weights of 50kg or more, the usual adult dosage should be used. Doses greater than 80mg/kg body weight should be avoided because of the increased risk of biliary precipitates.

Renal and hepatic impairment

In patients with impaired renal function, there is no need to reduce the dosage of hovid Hoftrex provided liver function is intact. Only in cases of pre-terminal renal failure (creatinine clearance < 10ml per minute) should the daily dosage be limited to 2g or less.

In patients with liver damage there is no need for the dosage to be reduced provided renal function is intact. In severe renal impairment accompanied by hepatic insufficiency, the plasma concentration of hovid Hoftrex should be determined at regular intervals and dosage adjusted.

In patients undergoing dialysis, no additional supplementary dosing is required following the dialysis. Serum concentrations should be monitored, however, to determine whether dosage adjustments are necessary, since the elimination rate in these patients may be reduced.

DIRECTIONS FOR USE

Do not use diluents containing calcium, such as Ringer's Solution or Hartmann's Solution, to reconstitute ceftriaxone. Particulate formation can result.

Amount of Sterile Water for Injection to be added during reconstitute

Vial Dosage Size	IV administration	IM administration
1 g	9.6 ml	3.6 ml

From a microbiological point of view, the product should be used immediately after reconstitution and any unused portion must be discarded.

Compatibility and stability

As a general rule, the solutions should be used immediately after preparation. Reconstituted Ceftriaxone for injection retain their potency for 6 hours if kept below 25°C and for 24 hours if refrigerated at 2 - 8°C. The solutions range in colour from pale yellow to amber, depending on the concentration and length of storage. The coloration of the solutions is of no significance for the efficacy or tolerance of the drug.

Note: The information given here is limited. For further information, consult your doctor or pharmacist.

Storage	: Store below 30°C. Protect from light and moisture.
Presentation/Packing	: 1 Vial Sterile Powder + 1 Ampoule Sterilized Water for Injection 10 ml. Packed in clear, transparent, glass vial with grey coloured bromobutyl rubber stopper and aluminium seal with coloured propylene disc.
Product Registration Holder	: HOVID Bhd.,121, Jalan Tunku Abdul Rahman, 30010 Ipoh, Malaysia.
Manufacturer	: Nectar Lifesciences Ltd Unit-VI, Village Bhatoli Kalan, (Adjoining Jharmajri E.P.I.P), P.O. Barotiwala, Tehsil Baddi, Distt. Solan (H.P.)-174103, India.
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